

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
AMTI – NMTC – **25th October, 2025** – INTER FINAL
Timings: 10am to 1pm

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Instructions:

1. Answer all questions. Each question carries 10 marks.
 2. Elegant and innovative solutions will get extra marks.
 3. Diagram and justification should be given wherever necessary.
 4. Before start answering, fill in the FACE SLIP completely.
 5. Your **Rough Work** should be in the answer sheet itself.
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1. Prove that

$$88 < 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{2024}} < 89$$

2. Given 11 real numbers in the form of infinite decimal representations, prove that there are two of them which coincide at infinitely many decimal places. Show that 11 is the smallest possible number that guarantees the existence of two such numbers.
3. In a triangle ABC , $AB = AC$ and I is the incenter of the triangle. The circle Γ_1 has center A and radius AB , and the circle Γ_2 has center I and radius IB . The circle Γ_3 passing through B and I intersects Γ_1 and Γ_2 at P and Q respectively (different from B). Let the lines IP and BQ intersect at R . Show that BR is perpendicular to CR .
4. Consider the sequence $1, 11, 111, 1111, \dots$ written in base 10 using only the digit 1. Find all primes p that divide infinitely many numbers in the sequence.
5. Let a, b, c be positive real numbers. Show that

$$\left(1 + \frac{a}{b}\right) \left(1 + \frac{b}{c}\right) \left(1 + \frac{c}{a}\right) \geq 2 \left(1 + \frac{a+b+c}{\sqrt[3]{abc}}\right)$$

6. Find all polynomials $P(x)$ of degree n such that $P(2) = 2$ and such that for all $x \in \mathbb{R}$,

$$P(1) + P(x) + P(x^2) + \cdots + P(x^n) = (1 + x + x^2 + \cdots + x^n)P(x)$$

7. Let m, n be positive integers. A total of $m + n$ lines are drawn in the plane such that

- (a) m lines are horizontal
- (b) Each of the other n lines intersect all the horizontal lines
- (c) No two of these n lines are parallel
- (d) No three of the $m + n$ drawn lines are concurrent

Find all possible values of the pair (m, n) if the number of regions created by these $m + n$ lines is 2022.

8. An acute angled triangle ABC is given. A circle passing through A and the circumcenter O of the triangle intersects AB and AC at points P and Q respectively. Prove that the orthocenter of the triangle POQ lies on the line BC .

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